

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination May 2017

PH109 Pharmaceutical Engineering - II

Date: 12.05.17, Friday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

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|---|--|----|
| A | Explain the theory of filtration. Write in detail about the factors affecting filtration. | 10 |
| B | a) Write a short note on cartridge filter. | 05 |
| | b) Write a short note on Membrane filter. | 05 |
| C | Explain in detail about Perforated Basket Centrifuge with proper diagram and applications of it. | 10 |

Q 2 Attempt any TWO of the following

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|---|--|----|
| A | Explain in detail about phase equilibria of evaporation with proper example. | 10 |
| B | Explain in detail about film evaporators. | 10 |
| C | Explain the following terms in brief. (Each carries 2 marks) | |
| | a) Centrifugation process | 02 |
| | b) Raoult's law | 02 |
| | c) McCabe Thiele method | 02 |
| | d) Microwave | 02 |
| | e) Relative humidity | 02 |

SECTION- II

Q 3 Attempt any TWO of the following

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|---|--|----|
| A | Explain by a detailed note on flash distillation. | 10 |
| B | Enumerate the difference between molecular and steam distillation. | 10 |
| C | Define drying. Explain the mechanism of drying. How it can be differentiated from evaporation? | 10 |

Q 4 Attempt any TWO of the following.

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|---|---|----|
| A | Write a detailed note on freeze dryer. | 10 |
| B | Write a detailed note on Refrigerator with proper diagram. | 10 |
| C | Explain in detail about dehumidification with proper diagram. | 10 |